

Lenovo Legion Go – Portable Unit for the Monkey Head Project

The **Lenovo Legion Go** is a handheld gaming PC that Lenovo introduced as a direct competitor to devices like the Steam Deck and Asus ROG Ally. It's essentially a full Windows 11 computer in a portable console form factor, and in the context of the Monkey Head Project, it will serve as a **flexible, modular "Huey" unit** – i.e. a portable assistant on the go. What makes the Legion Go particularly notable (besides its powerful internals) is its **unique detachable controller design**, which adds versatility in how it can be used within the project. Below, we break down the Legion Go's key specifications, features, and how it can be integrated and optimized for performance in the project's use-case.

Key Hardware Specifications and Features

- **Processor & Graphics:** AMD Ryzen **Z1 Extreme** 8-core/16-thread APU (up to ~5.1 GHz boost) with **12 RDNA 3** graphics compute units (integrated GPU up to 2.7 GHz) ¹. This is the same high-end chip used in the ROG Ally, offering comparable performance to a mid-range gaming PC in handheld form ². (The integrated GPU shares system memory and is roughly equivalent to a dedicated graphics card in the ~3GB VRAM class, suitable for 720p to 1080p gaming).
- **Memory:** 16 GB of **LPDDR5X-7500** RAM (soldered) ³, providing ample memory bandwidth for both the CPU and integrated GPU.
- **Storage:** **1 TB PCIe Gen4 NVMe SSD** (M.2 2242 form factor) ⁴ for fast storage. This is a generous amount of onboard storage for the operating system, project files, and even games. Additionally, the Legion Go has a **microSD card slot** for expansion – it supports cards up to 2 TB, which can more than double the total storage capacity for data or media ⁵.
- **Display:** 8.8-inch **WQXGA** touchscreen (2560×1600 resolution, 16:10 aspect ratio) at up to **144 Hz** refresh rate ¹. This IPS panel is bright (500 nits) and protected by Gorilla Glass 5 ¹. The high resolution and refresh rate are standout features, though running at lower resolutions (like 1200p or 800p) is often preferable for better performance and battery life.
- **Detachable Controllers:** Two gamepad controllers attach to the sides of the screen (similar to Nintendo Switch Joy-Cons). These controllers can be **released via hidden latches** and used wirelessly when detached ⁶. They include dual analog sticks (hall-effect sensors to prevent drift), D-pad, ABXY buttons, analog triggers, bumpers, a **small trackpad** on the right controller, a scroll wheel, and **six assignable grip buttons** on the back ⁷ ⁸. *(More on the controllers in the next section.)*
- **Battery:** 49.2 Wh rechargeable battery ⁹. This capacity is slightly larger than Steam Deck/ROG Ally's ~40 Wh batteries, but the powerful hardware means **battery life is still limited** – in heavy use (gaming at high settings) it can drop under 1 hour, while more moderate use or power-saving settings can yield around 2–3 hours on a charge ¹⁰. (In best-case scenarios with tweaks, testers have managed roughly 3 hours, which is **better than the Ally but shorter than the Steam Deck** under similar conditions ¹⁰.)
- **Ports & Connectivity:** Uncommonly for a handheld, the Legion Go offers **two USB-C 4.0 ports** (one on top, one on the bottom) ¹¹. Both support USB4/Thunderbolt-like speeds and DisplayPort 1.4

output, as well as Power Delivery charging (65W). This dual-port design means you can charge the device and connect peripherals or an external display simultaneously without a dock ¹¹. There's also a 3.5mm audio jack for headphones ¹² and **Wi-Fi 6E + Bluetooth 5.2** wireless connectivity ¹².

- **Operating System:** Ships with **Windows 11** (and in this case is running the Windows 11 Insider Preview). This provides full PC functionality and software compatibility – important for integrating the Monkey Head Project's software stack, which is designed to be cross-platform ¹³.

Overall, the Legion Go's hardware is top-tier for a handheld device. It essentially packs ultrabook-level internals into a portable form. Next, we'll look at the Legion Go's signature feature – its **detachable controllers** – and why that matters for modular use.

Unique Detachable Controllers and Modular Design

One defining feature of the Lenovo Legion Go is its **modular controller system**. The two controllers can slide on and off the sides of the main unit, very much like a Nintendo Switch's Joy-Cons, but Lenovo's mechanism is more robust and **well-engineered with minimal wobble** when attached ¹⁴ ¹⁵. You simply **press a release switch and slide them off**, and they'll automatically connect via Bluetooth to the main unit when detached ¹⁶. The device also has a wide, sturdy **kickstand** on the back, spanning almost the entire width of the unit, so you can prop the screen up on a table when using the controllers off-device ¹⁶.

Each controller is feature-packed: the **right-hand controller** includes a tiny circular trackpad (useful for mouse cursor movement in Windows or strategy games) and even a **scroll wheel**, which is uncommon on game controllers ⁸. Notably, the **bottom of the right controller has an optical sensor and a switch for "FPS mode"** ¹⁷. In FPS mode, you can attach a small included cover to the controller that lets you use it like a **vertical mouse** – the optical sensor tracks movement when you slide the controller on a surface, and the triggers act as mouse buttons ¹⁷ ¹⁸. This effectively turns the Legion Go into a mini tabletop PC: one controller becomes a mouse and the other can be used like a joystick or gamepad in the other hand. This mode is ideal for precision aiming in shooters or navigating Windows interface when docked on a table.

Furthermore, both controllers have **haptic feedback** and six-axis gyros for motion controls ¹⁸, and Lenovo used **hall-effect joysticks** to avoid drift issues over time ¹⁹. There are also extra **grip buttons (M1, M2, M3, Y1, Y2, Y3)** on the backs of the controllers that can be custom-mapped for additional functions ⁷, which could be useful for custom commands in the project's applications.

From a modularity perspective, this detachable design means the Legion Go can **serve multiple roles**: it's a handheld console when you need all-in-one portability, but it can also act like a small desktop or tablet when you set it down and remove the controllers. For the Monkey Head Project, this flexibility is a big plus – you could, for example, mount the main tablet portion onto a robotic frame or a stand (as a display or control unit) while an operator uses the wireless controllers to interact from a short distance. The **kickstand** supporting various angles makes it practical to deploy the device on a desk or lab bench and still comfortably use it. In short, the Legion Go's hardware isn't just powerful; it's **reconfigurable**, which aligns well with the project's need for a adaptable, modular assistant unit.

Integration as a Portable "Huey" Unit in the Project

Within the Monkey Head Project, *Huey* refers to the **robotic shell or physical embodiment** of the GenCore AI Operating System ²⁰. In the full lab setup, Huey is a large, stationary unit with high-end hardware (like

server-grade motherboards, etc.) to run the AI and interface with the world. The Lenovo Legion Go offers the opportunity to have a **portable mini-Huey** – essentially a scaled-down version of the AI’s hardware platform that can be carried around by a person or placed on a mobile robotic base.

Despite its much smaller size, the Legion Go’s specs (discussed above) are quite formidable. The **8-core Z1 Extreme CPU and 16GB RAM** mean it can run the GenCore AIOS and related Monkey Head Project software without needing a full desktop tower. In fact, the project documentation emphasizes *broad compatibility* with multiple operating systems and hardware configurations, aiming to be “universally accessible” ¹³ . Having Windows 11 on the Legion Go is not an issue, as the GenCore AI and its subsystems are designed to integrate with Windows environments alongside Linux or macOS ¹³ . This means the AI services, sensor interfaces, and user interaction modules can all run on the Legion Go when it’s deployed as a field unit.

The **Legion Go is listed as one of the development platforms** for the project (alongside a MacBook Pro, etc.) ²¹ , which confirms that the team is actively using it to test and run the AI system. As a portable Huey, the Legion Go could be used by a lab assistant or engineer on the move – for instance, carrying it to different lab stations or even off-site. It can function as a standalone AI assistant: the user can speak to it (through a connected headset or external microphone, since the device itself doesn’t have built-in mics/cameras like a phone), view data on its screen, and even connect AR/VR goggles or other peripherals if needed. In fact, Lenovo has an accessory called Legion Glasses (AR glasses) that can pair with the Legion Go for heads-up display, pointing to future **AR integration** possibilities ²² ²³ .

Thanks to the **dual USB4 ports**, integration into larger setups is straightforward. For example, in a lab one could plug the Legion Go into a power source and connect it to external devices: **monitors, keyboards/mice, or even an external GPU** if extreme graphics performance were needed. The USB-C ports support DisplayPort-out and power delivery simultaneously, so the Legion Go can be docked and act almost like a small desktop when the situation calls for it ¹¹ . This could be useful if the AI needs to display a lot of information on a big screen during a demonstration, or if it needs to interface with specialized hardware (via USB adapters) while still being easily removable and portable.

In summary, using the Lenovo Legion Go as a portable Huey unit provides the Monkey Head Project with a **highly versatile, mobile platform** for their AI. It bridges the gap between the fixed, powerful lab hardware and a truly mobile assistant. While it doesn’t match the raw power of the full-sized Huey rig (with server-grade components), it offers sufficient performance to run the AI in the field, and its modular form allows it to adapt to different scenarios (handheld use, table-top use with detached controllers, or docked use in a larger setup).

Performance Profile and Battery Life

Given its powerful AMD APU, the Legion Go’s performance is impressive for a handheld. In practical terms, it can run modern 3D games and heavy applications – an important consideration if the Monkey Head Project’s AI involves real-time processing or even some on-device machine learning inference. The **Ryzen Z1 Extreme** is essentially on par with an AMD Ryzen 7 7840U (8-core Zen4 architecture), and the integrated RDNA3 GPU (similar to Radeon 780M graphics) can handle many games at respectable settings. Benchmarks in reviews have shown that at **20W TDP (Performance mode)** – which is the default max on battery – the Legion Go achieves, for example, around **64 FPS in Doom Eternal (Medium settings, at ~1200p)** and around **80 FPS in Star Wars: Squadrons (Low settings)** ²⁴ . This indicates the device can comfortably handle 3D environments or GUI-intensive tasks the project might throw at it, especially if the

resolution or fidelity is tuned for performance. In everyday terms, any productivity, data processing, or moderate AI tasks (like running large language model prompts, computer vision inference with optimized models, etc.) should run smoothly on this device.

One thing to note is that the Legion Go's high native screen resolution (2560×1600) can be very demanding. For optimal performance (and to conserve energy), the **project team will likely run applications at a lower resolution** or use resolution scaling when ultra-high detail isn't needed. The 144 Hz refresh capability is mainly beneficial for gaming; for AI assistant use, capping the refresh rate to 60 Hz can also help with battery life without much downside.

On the topic of **battery life**, as mentioned, the Legion Go can be quite power-hungry. The 49 Wh battery is physically larger than many competitors', yet under heavy load (gaming, stress tests) users have reported **draining the battery in under an hour** ²⁵. For less intense usage (web browsing, coding, running simpler AI tasks), you might get a few hours. In direct comparisons, the Legion Go tends to last **a bit longer than the Asus ROG Ally** (which has a smaller battery) but still **significantly less than a Steam Deck** running Linux, which is heavily optimized for efficiency ¹⁰. One review noted that after Valve's optimizations, a Steam Deck could reach 5 hours of light use, whereas the Legion Go **"gets 3 at best** with the same tweaks" ¹⁰.

However, for the Monkey Head Project's use as a portable assistant, there are ways to manage this: running in **"Balanced" or "Quiet" power modes** for routine tasks can cap the APU to a lower wattage (e.g., 8-15W), dramatically extending runtime (at the cost of some performance) ²⁶. The device has a physical mode switch (with an LED indicator: blue for Quiet, white for Balanced, red for Performance) to quickly toggle these modes ²⁷. This means a user can dynamically choose to conserve battery when doing light tasks (like note-taking or idle listening) and then kick into Performance mode (20W TDP) when heavy computation is needed. Additionally, because the Legion Go is a full Windows PC, traditional power management (Windows battery saver, dimming screen, etc.) applies as well. And of course, whenever possible, using the device **while plugged in** (or carrying a USB-C power bank) can alleviate runtime concerns.

Thermal performance of the Legion Go is quite good for its class. Thanks to a robust cooling solution (with ample ventilation and a large fan), it tends to **stay relatively cool to the touch under load** ²⁸ – an important consideration if someone will hold it or have it close to them while it's working. Reviewers observed internal temperatures in the 60°C range on Performance mode, and surface temperatures that were warm but not uncomfortable ²⁹ ³⁰. The fan can get audible (running up to ~34 dB(A) in Performance mode) ³¹, but in many cases, especially if running lighter workloads or when the fan is in Quiet mode, noise should not be a major distraction. For an AI assistant usage, this means the device likely won't overheat even during sustained computations, and any fan noise is a reasonable trade-off for keeping the device cool.

In summary, **out-of-the-box performance** of the Legion Go should be more than sufficient for a portable Monkey Head Project unit. The key is balancing that performance with battery life and heat – using the right performance profile for the task at hand. In the next section, we'll discuss potential **modifications and tweaks** to further improve performance, efficiency, and overall suitability of the Legion Go for the project.

Potential Modifications and Performance Improvements

To get the most out of the Lenovo Legion Go as a project platform, there are several modifications and optimizations that can be considered. These range from simple software settings to more involved hardware tweaks:

- **TDP and Power Settings:** The Legion Go comes with **built-in power profiles** (accessed via Lenovo's Legion Space software or a quick sidebar) that let you adjust the APU's TDP on the fly. For instance, Quiet mode limits wattage (for cooler, quieter operation), while Performance mode targets ~20W on battery ³². There's also a Manual mode where users can attempt higher TDP (25W or 30W) when plugged in. However, testing shows going beyond ~20W on battery yields **diminishing returns** – in fact, one review noted that switching from 20W to 30W while on battery sometimes **worsened performance** due to power or thermal constraints ³³. The takeaway: using the built-in modes is usually effective, and **tuning TDP** to the task (lower for efficiency, higher for heavy computation when plugged in) can optimize performance per watt. The quick-toggle button makes this easy to manage on the go.
- **Undervolting the APU:** If squeezing out more efficiency is desired, **undervolting** is an advanced tweak. By slightly lowering the voltage to the CPU/GPU while maintaining clock speeds, you reduce heat and power draw, which can either prolong battery life or allow higher clocks within the same power budget. Officially, Lenovo doesn't advertise undervolting, but some users have reported that the Legion Go's BIOS has options for it, and tools like the **Universal X86 Tuning Utility** can be used in Windows to undervolt the Z1 Extreme ³⁴. This isn't a necessity for most users – as one Legion Go owner noted, you might not *need* to undervolt since you can just use the TDP slider to manage performance/power ³⁵. But for enthusiasts in the project, a stable undervolt could be beneficial (e.g., it might shave off a few degrees of heat and extend battery life during intensive computations). Caution is advised: every chip is different, so any undervolt should be tested thoroughly for stability.
- **Thermal Paste Upgrade:** Improving the cooling interface is another tweak that's been tried. The stock thermal paste on the APU can be replaced with a higher-performance compound. Community reports indicate that using a quality paste like **Arctic MX-4 can noticeably lower temperatures**, whereas even more exotic solutions like Honeywell's PTM7950 phase-change pad gave only modest gains in this particular device ³⁶. Re-pasting the device requires opening it up (which is a delicate process), so it's an optional mod for those comfortable with hardware. If done properly, better cooling can help the Legion Go sustain high performance longer (less throttling) and possibly make the fan quieter under load. Given that the Legion Go already stays fairly cool externally, this mod is more about maximizing internal performance and longevity.
- **Screen and Refresh Rate Tuning:** As mentioned earlier, running the display at full 2560×1600 and 144 Hz is overkill for many use cases and drains power. For practical use, especially in the project context, one can **reduce the resolution or use Windows scaling** to effectively render at a lower resolution (like 1920×1200 or 1280×800) when the ultra-fine detail isn't needed. This can boost frame rates significantly (important if the AI interface has any 3D visualization or if the device is used for demos/gaming) ³⁷. Similarly, setting the refresh rate to 60 Hz except when 144 Hz is specifically needed will help both battery and thermals. These are simple software settings that can be switched as needed.

- **Storage and Memory Upgrades:** The 1TB NVMe SSD is quite fast (Gen4 speeds) and more than sufficient for most uses. But since it's a standard M.2 2242 drive, it is **technically replaceable** if a larger capacity or a faster model is desired in the future ⁴. For example, one could install a 2TB 2242 SSD to have more onboard storage for AI datasets or media. The microSD slot, as mentioned, offers up to an additional 2TB of removable storage ⁵ – perfect for keeping large, less frequently used files or backups. The RAM, however, is soldered and not upgradable; 16GB is the fixed amount. If the project workflows ever require more than 16GB RAM (for running multiple AI models simultaneously, etc.), swapping to a device with more memory might be the only solution – but 16GB is a generous amount for a mobile unit and aligns with the project's current design (the MacBook Pro 2019 dev machine likely has a similar RAM size, for reference).
- **External GPU (eGPU) and Docking:** Thanks to the **dual USB4 ports**, the Legion Go has the uncommon ability (for a handheld) to connect to external PCIe devices. This means in a stationary setting, one could attach an **eGPU enclosure** via USB4/Thunderbolt, allowing a desktop-class GPU to drive graphics or GPU-accelerated computations. This is more of a hypothetical enhancement – it hasn't been widely reported in reviews – but USB4 theoretically supports Thunderbolt 3/4 devices, so an eGPU **might** work (bandwidth would be limited compared to internal GPU slots, but still a huge boost for tasks like 3D rendering or machine learning that can use a powerful GPU). Even without an eGPU, the ports allow connecting to multiple external displays and peripherals easily. Lenovo specifically positioned one USB-C on the top and one on the bottom so that docking and charging scenarios are very flexible ¹¹. For example, you can prop the Legion Go on its kickstand, plug in power to the bottom USB-C, and plug the top USB-C into a monitor or hub – turning it into a makeshift desktop. For the project, this means during development or presentations, the Legion Go can act as a computer powering large screens or specialized equipment, and then be quickly **undocked and taken on the go**.
- **Software Updates and OS Optimization:** Since the unit is running **Windows 11 Insider Preview**, it's likely receiving the latest fixes and features for handheld devices. Microsoft has been improving Windows support for devices like these (e.g. better on-screen keyboard, UI scaling, game mode improvements for handhelds). Keeping the OS updated will ensure any **early bugs** (and Lenovo's own software bugs) get patched – early reviews noted that the Legion Go's software (the Legion Space app, etc.) still had some issues at launch ³⁸. It's wise to update Lenovo's drivers/firmware as well, as they have already pushed a few updates to refine fan curves and controller behavior. Additionally, for project-specific optimization, one can **trim unnecessary background apps** on Windows, use power profiles, and ensure that the AI software services start on boot and are configured for a touchscreen interface, etc., to make the device as user-friendly and responsive as possible in the field.

In summary, the Legion Go is quite capable out of the box, but the above tweaks can tailor its performance and functionality to better suit the Monkey Head Project's needs. Adjusting power settings gives control over performance vs. battery life, undervolting and improved cooling can eke out more sustained performance, and the hardware's inherent expandability (ports/storage) provides avenues to integrate with larger setups or handle more data. All these modifications and improvements ensure that the Legion Go can reliably function as a **portable AI hub** – delivering high performance when required and conserving energy when portability is paramount.

References: The information above was gathered from Lenovo Legion Go hardware reviews and specifications, as well as the Monkey Head Project documentation for context. Key sources include the Lenovo Legion Go spec sheet and hands-on reports ¹ ¹¹, an in-depth review by UltrabookReview ⁶ ²⁴ ¹⁰, Tom's Hardware's first-look article ¹⁶ ¹⁸, and community discussions by early Legion Go users ³⁵ ³⁶. These sources corroborate the Legion Go's capabilities (e.g. its detachable controller design, performance metrics, and battery life) and provide insight into potential enhancements for using it as a mobile project unit. The Monkey Head Project's README was also referenced to ensure alignment of this device's use with the project's goals and compatibility ²⁰ ¹³ ²¹.

¹ ² ³ ⁴ ⁶ ⁷ ⁹ ¹⁰ ¹² ¹⁴ ¹⁵ ¹⁷ ²² ²³ ²⁴ ²⁵ ²⁶ ²⁷ ²⁸ ²⁹ ³⁰ ³¹ ³² ³³ ³⁷ ³⁸ **Lenovo Legion Go review (AMD Z1 Extreme)**

<https://www.ultrabookreview.com/66707-lenovo-legion-go-review/>

⁵ ⁸ ¹¹ ¹⁶ ¹⁸ ¹⁹ **Lenovo Legion Go Brings Ryzen Z1 Extreme Power to a Switch Design | Tom's Hardware**

<https://www.tomshardware.com/news/lenovo-legion-go-hands-on-price-specs>

¹³ ²⁰ ²¹ **README.md**

<https://github.com/DylanLRPollock/Monkey-Head-Project/blob/19807780854c129334f74dfd66be6e39e9b07f55/README.md>

³⁴ ³⁵ **Undervolting on LEGION GO : r/LegionGo**

https://www.reddit.com/r/LegionGo/comments/1fqdbkm/undervolting_on_legion_go/

³⁶ **PTM7950 on the Go : r/LegionGo - Reddit**

https://www.reddit.com/r/LegionGo/comments/1801i8u/ptm7950_on_the_go/